

Wilcoxon Signed Ranks test.

KEEL non-parametric statistical module

March 1, 2025

	(1)	(2)	(3)
GWO (1)	-	78066.0	74276.0
SA (2)	540.0	-	219.0
HC (3)	4330.0	78387.0	-

Table 1: Ranks computed by the Wilcoxon test

	(1)	(2)	(3)
GWO (1)	-	●	●
SA (2)	○	-	○
HC (3)	○	●	-

Table 2: Summary of the Wilcoxon test. ●= the method in the row improves the method of the column. ○= the method in the column improves the method of the row. Upper diagonal of level significance $\alpha = 0.9$, Lower diagonal level of significance $\alpha = 0.95$

	$\alpha = 0.9$		$\alpha = 0.95$	
Method	+	$\pm$	+	$\pm$
GWO	2	2	2	2
SA	0	0	0	0
HC	1	1	1	1

Table 3: Wilcoxon test summary results